Thursday Warm-up: Potentials

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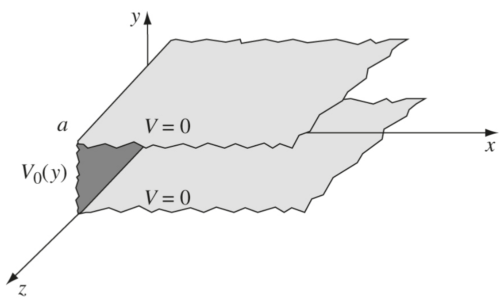

Figure 1: An infinite "slot" in the xy-plane, defined by two grounded conductors in the xz-plane separated by a distance a, with boundary condition $V_0(y)$ in the yz-plane.

1 Memory Bank

- **Fourier series** for a function $f(x)$:

$$f(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} (A_n \cos nx + B_n \sin nx) \quad (1)$$

$$A_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos(nx)dx \quad (n = 0, 1, 2, ...) \quad (2)$$

$$B_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin(nx)dx \quad (n = 1, 2, ...) \quad (3)$$

- Laplace's equation in 2D Cartesian coordinates:

$$\frac{\partial^2 X}{\partial x^2} + \frac{\partial^2 Y}{\partial y^2} = 0 \quad (4)$$

2 Separation of Variables

1. **Fourier Series**. Suppose a function is $f(x) = x$ between 0 and 2π, and then $f(x) = (x - 2\pi)$ from 2π to 4π, and then $f(x) = (x - 4\pi)$ from 4π to 6π, ..., repeating periodically. (a) Develop the *Fourier series* for $f(x)$ by computing the coefficients A_n and B_n using equations in the Memory Bank. (b) Equations 2 and 3 impose *boundary conditions* that build Eq. 1. (c) Using software, plot the series up to $n = 5$.

2. Two infinite grounded metal plates lie parallel to the xz plane, one at $y = 0$, and the other at $y = a$ (Fig. 1). The left end, at $x = 0$, is closed off with an infinite strip insulated from the two plates, and maintained at a voltage $V_0(y)$. Find the voltage inside the slot.

   - Write down *four* boundary values as equations.

   - Let the solution be *separable*:

   $$V(x, y) = X(x)Y(y) \quad (5)$$

     Substitute this into the 2D version of Laplace's equation in Cartesian coordinates, and separate the X and Y portions into distinct terms.

   - Argue that each term should be a *constant*, for if (for example) y is held constant and x is varied, the left side of the equation would vary, while the right side would be constant at 0.
   - Since the X constant and Y constant are equal and opposite, set the X constant to k^2 and the Y constant to $-k^2$. Try to guess the solutions for the separated equations for $X(x)$ and $Y(y)$.

   - Impose the boundary conditions for $x \neq 0$, and write an example solution for $V(x, y)$.

   - Laplace's equation admits *linear combinations* of solutions: $V = a_1V_1 + a_2V_2 + ...$ Write a series of solutions based on your example solution.